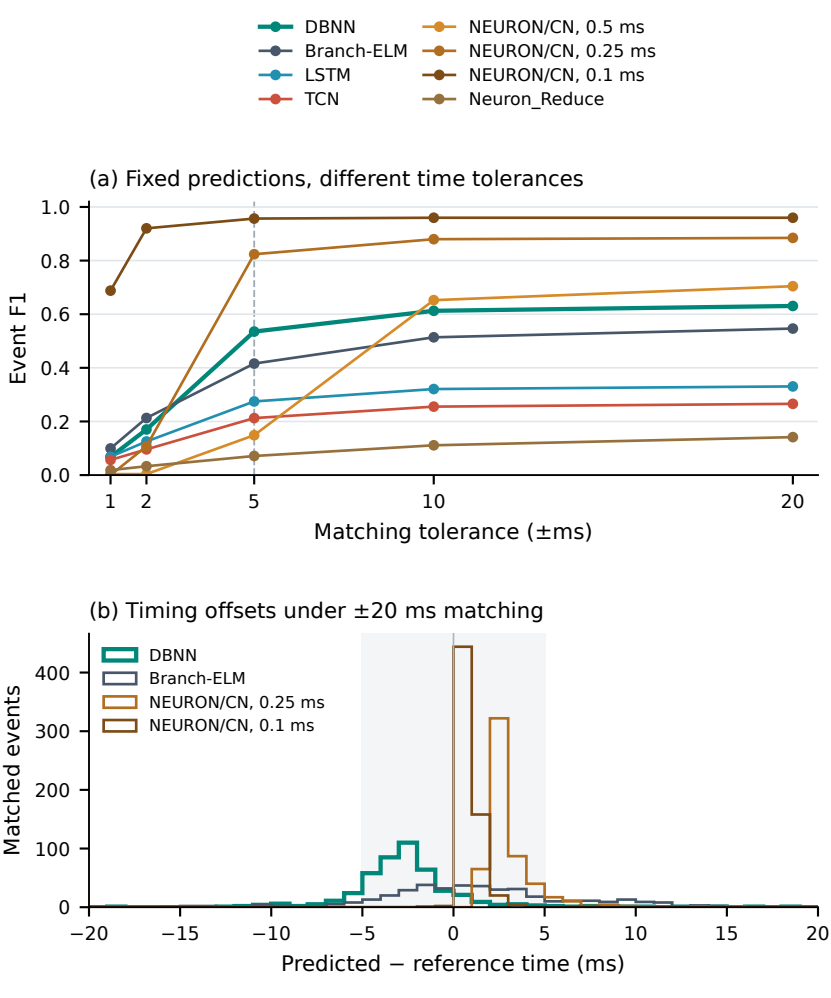
DBNN
Branch-ELM
LSTM
TCN
NEURON/CN, 0.5 ms
NEURON/CN, 0.25 ms
NEURON/CN, 0.1 ms
Neuron_Reduce
(a) Fixed predictions, different time tolerances
Event F1
1.0
0.8
0.6
0.4
0.2
0.0
1
2
5
10
20
Matching tolerance (±ms)
(b) Timing offsets under ±20 ms matching
DBNN
Branch-ELM
NEURON/CN, 0.25 ms
NEURON/CN, 0.1 ms
Matched events
400
300
200
100
0
−20
−15
−10
−5
0
5
10
15
20
Predicted − reference time (ms)